MCC Entrance Test, August 2026 — Day 1 — Solutions

Level T3

Test Information. Level T3 — 150 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra (×2), Geometry (×2), Combinatorics, Number Theory.
Rules: show your work (partial credit rewards genuine progress, but claims still need real justification at this level); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts).[1] Two positive integers x and y satisfy $xy + x + y = 71$ and $x^2y + xy^2 = 880$. Find $x^2 + y^2$.

Official solution. Let $s = x + y$ and $p = xy$. The first equation gives $p + s = 71$. The second equation factors as $xy(x + y) = ps = 880$. So s and p are roots of

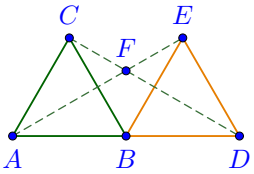
$$t^2 - 71t + 880 = 0 \implies (t - 16)(t - 55) = 0 \implies \{s, p\} = \{16, 55\}.$$

Since x, y are positive integers, $s^2 - 4p$ must be a perfect square, so that the quadratic $u^2 - su + p = 0$ they satisfy has integer roots.

Checking $s = 55,\ p = 16$: $s^2 - 4p = 3025 - 64 = 2961$, which is not a perfect square ($54^2 < 2961 < 55^2$).

Checking $s = 16,\ p = 55$: $s^2 - 4p = 256 - 220 = 36 = 6^2$, giving $u = \frac{16 \pm 6}{2} = 11, 5$ – integers. $\{s, p\} = \{16, 55\}$, and $x^2 + y^2 = s^2 - 2p = 16^2 - 2(55) = 256 - 110 = \boxed{146}$. □

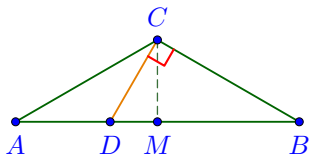
Problem 2 (10 pts).[2] Two congruent equilateral triangles ABC and BDE are positioned so that A, B, D lie on a single line, in that order, with C and E on the same side of this line. Let F be the intersection point of segments CD and AE. Find $\angle AFD$.



Official solution. Since $\triangle ABC$ is equilateral, $AB = CB$ and $\angle ABC = 60°$; likewise $\triangle BDE$ equilateral gives $BE = BD$ and $\angle DBE = 60°$. As A, B, D are collinear with B between them, $\angle ABD = 180°$, so $\angle ABE = \angle ABD - \angle DBE = 180° - 60° = 120°$, $\angle CBD = \angle ABD - \angle ABC = 180° - 60° = 120°$.

$\angle ABE = \angle CBD$, $AB = CB$, and $BE = BD$, so $\triangle ABE \cong \triangle CBD$ by SAS. Let $x = \angle BAE = \angle BCD$ and $y = \angle AEB = \angle CDB$, the angle sum in $\triangle ABE$ gives $x + 120° + y = 180°$, so $x + y = 60°$.

Now consider $\triangle AFD$. Since D lies on ray AB beyond B, $\angle FAD = \angle FAB = \angle BAE = x$; since A lies on ray DB beyond B, $\angle FDA = \angle FDB = \angle CDB = y$. Hence $\angle AFD = 180° - x - y = 180° - 60° = \boxed{120°}$. □



Problem 3 (15 pts).[3] Triangle ABC has $AC = BC$, $AB = 3$. Point D lies on side AB so that $BD = 2\,AD$. $\angle BCD = 90°$, find $\angle BAC$.

(a) M is the midpoint of AB. Prove that $CM^2 = DM \cdot MB$. [9 pts]

(b) Using part (a), find $\angle BAC$. [6 pts]

Official solution. **Part (a).** Since $AC = BC$, triangle ABC is isosceles with apex C, so the altitude from C bisects the base: $CM \perp AB$ with M the midpoint of AB.

Since $AD : DB = 1 : 2$, D sits at $\frac{1}{3}AB$ from A, while M sits at $\frac{1}{2}AB$ from A; since $\frac{1}{3}AB < \frac{1}{2}AB < AB$ for any positive length AB, the order A, D, M, B holds along the segment, so M lies on segment AB between D and B. Hence CM is also the altitude from the right angle C of $\triangle BCD$ down to its hypotenuse BD. By the geometric-mean (altitude-on-hypotenuse) relation, $CM^2 = DM \cdot MB$.

Now $AD = 1$, $DB = 2$, $AM = MB = 1.5$, so $DM = AM - AD = 0.5$. Thus $CM^2 = (0.5)(1.5) = \boxed{0.75}$.

Part (b). In right triangle ACM (right angle at M), the Pythagorean theorem gives

$$AC^2 = AM^2 + CM^2 = 1.5^2 + 0.75 = 2.25 + 0.75 = 3,\ AC = \sqrt{3}.$$

So $CM : AM : AC = \frac{\sqrt{3}}{2} : \frac{3}{2} : \sqrt{3} = 1 : \sqrt{3} : 2$ – the side ratio of a 30-60-90 triangle, with CM (the side opposite $\angle BAC$) playing the role of the shortest side. Hence $\angle BAC = \boxed{30°}$. □

[1] American Invitational Mathematics Examination (AIME). [2] Matematická Olympiáda (Czech Mathematical Olympiad), Category Z. [3] Olimpiada Matematyczna Juniorow (Polish Junior Mathematical Olympiad).

Problem 4 (15 pts).[4] Let a and b be real numbers with $a > b$, satisfying

$$2^a + 2^b = 75 \qquad \text{and} \qquad 2^{-a} + 2^{-b} = \tfrac{1}{12}.$$

(a) Let $x = 2^a$ and $y = 2^b$. Find xy. [6 pts]

(b) Solve for x and y, then compute 2^{a-b}. [9 pts]

Official solution. **Part (a).** With $x = 2^a$, $y = 2^b$, the two conditions become

$$x + y = 75,\ \tfrac{1}{x} + \tfrac{1}{y} = \tfrac{1}{12} \implies \tfrac{x+y}{xy} = \tfrac{1}{12} \implies \frac{75}{xy} = \frac{1}{12} \implies xy = \boxed{900}.$$

Part (b). Since $x + y = 75$ and $xy = 900$, x and y are the roots of $t^2 - 75t + 900 = 0$.

$$75^2 - 4(900) = 5625 - 3600 = 2025 = 45^2 \implies t = \tfrac{75 \pm 45}{2} \implies t = 60 \text{ or } t = 15$$

$$a > b \implies x = 2^a > 2^b = y \implies x = 60,\ y = 15 \implies 2^{a-b} = \frac{x}{y} = \frac{60}{15} = \boxed{4}.$$

□

Problem 5 (25 pts).[0] Nine tokens are arranged in a row: three Copper tokens, three Iron tokens, and three Silver tokens (tokens of the same type are indistinguishable from one another). Call an arrangement *bad* if some three *consecutive* tokens are all the same type.

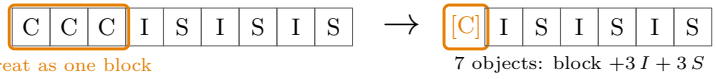
(a) How many distinct arrangements of the nine tokens are there, with no restriction? [4 pts]

(b) How many of those arrangements have all three Copper tokens consecutive (forming a single block), regardless of how the rest are arranged? [5 pts]

(c) Find the number of arrangements that are *not* bad. [16 pts]

Official solution. **Part (a).** The nine tokens consist of three groups of three identical tokens each, so the number of distinct arrangements is the multinomial coefficient $\dfrac{9!}{3!\,3!\,3!} = \boxed{1680}$.

Part (b). Treat the three Copper tokens as a single block. Together with the three Iron and three



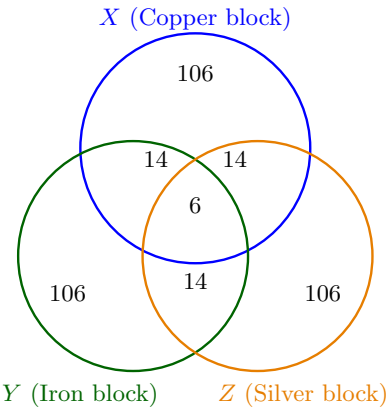
Silver tokens, this gives 7 objects to arrange (the block, plus $3 + 3$ identical tokens): $\dfrac{7!}{3!\,3!} = \boxed{140}$.

Part (c). Let X, Y, Z be the sets of arrangements (from part (a)'s 1680 total) in which the Copper, Iron, and Silver tokens, respectively, are all consecutive; since each token type has exactly three copies, "three consecutive tokens of the same type" happens exactly when that whole type forms one block, so the bad arrangements are exactly $X \cup Y \cup Z$. By part (b),

$$|X| = |Y| = |Z| = 140 \implies |X| + |Y| + |Z| = 420.$$

For two token types consecutive at once (e.g. $X \cap Y$): treat both as blocks, leaving 5 objects (2 blocks + 3 identical remaining tokens), giving $\frac{5!}{3!} = 20$ arrangements; there are 3 such pairs:

$$|X \cap Y| + |X \cap Z| + |Y \cap Z| = 3(20) = 60.$$



[4] Singapore Mathematical Olympiad.

For all three types consecutive at once: 3 distinguishable blocks in a row, giving $3! = 6$ arrangements. By inclusion–exclusion (elements in exactly two sets were added twice above and must be subtracted once; elements in all three sets were added three times and subtracted three times, so must be added back once),

$$|X \cup Y \cup Z| = 420 - 60 + 6 = 366.$$

The number of desired arrangements is $1680 - 366 = \boxed{1314}$. □

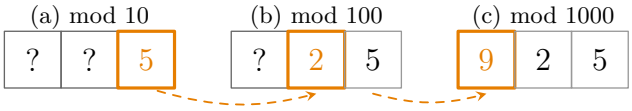
Problem 6 (25 pts).[0] A positive integer n is such that when the multiplier 2013 is applied – that is, the product $2013n$ is computed – the last four digits of the result are 2025. Find the sum of the digits of the *smallest* such positive integer n.

(a) Show that $2013n \equiv 2025 \pmod{10000}$, and find the last (units) digit of n. [8 pts]

(b) Find the tens digit of n. [5 pts]

(c) Continue this process to find the smallest positive integer n, then compute the sum of its digits. [12 pts]

The three parts below pin down the digits of n one at a time, starting from the units place:



each part pins down one more digit of n, from the units place leftward –
part (a) finds the units digit 5, part (b) the tens digit 2, part (c) the hundreds digit 9

Official solution. **Part (a).** "Last four digits are 2025" means $2013n \equiv 2025 \pmod{10000}$. Reducing modulo 10: $2013 \equiv 3$ and $2025 \equiv 5 \pmod{10}$, so $3n \equiv 5 \pmod{10}$.

Since $3 \times 7 \equiv 1 \pmod{10}$, multiplying both sides by 7: $n \equiv 35 \equiv \boxed{5} \pmod{10}$ – so n ends in the digit 5.

Part (b). Write $n = 10k + 5$. Then $2013n = 2013(10k + 5) = 20130k + 10065$. Modulo 100: $20130k \equiv 30k$ and $10065 \equiv 65$, so $2013n \equiv 30k + 65 \pmod{100}$. Since the last two digits of 2025 are 25,

$$30k + 65 \equiv 25 \pmod{100} \implies 30k \equiv -40 \equiv 60 \pmod{100}.$$

$$\implies 3k \equiv 6 \pmod{10} \implies k \equiv \boxed{2} \pmod{10}.$$

So $n = 10(10m + 2) + 5 = 100m + 25$ for some integer $m \geq 0$ – n ends in the digits 25.

Part (c). Write $n = 100m + 25$. Then $2013n = 2013(100m + 25) = 201300m + 50325$. Modulo 1000: $201300m \equiv 300m$ and $50325 \equiv 325$, so $2013n \equiv 300m + 325 \pmod{1000}$. Since the last three digits of 2025 are 025,

$$300m + 325 \equiv 25 \pmod{1000} \implies 300m \equiv -300 \equiv 700 \pmod{1000}$$
$$\implies 3m \equiv 7 \pmod{10} \implies m \equiv 9 \pmod{10}.$$

So $m = 10d + 9$ for some integer $d \geq 0$, giving $n = 100(10d + 9) + 25 = 1000d + 925$ – n ends in the digits 925. It remains to pin down d: reducing modulo 10000,

$$2013n = 2013(1000d + 925) = 2013000d + 1862025.$$

Since $2013000 \equiv 3000$ and $1862025 \equiv 2025 \pmod{10000}$, this gives $2013n \equiv 3000d + 2025 \pmod{10000}$ – already exactly the target residue 2025 when $3000d \equiv 0 \pmod{10000}$. Dividing through by $\gcd(3000, 10000) = 1000$, this is $3d \equiv 0 \pmod{10}$, and since $\gcd(3, 10) = 1$, $d \equiv 0 \pmod{10}$. The smallest non-negative choice is $d = 0$, giving the smallest positive integer $n = 1000(0) + 925 = \boxed{925}$.

Check: $2013 \times 925 = 1{,}862{,}025$, whose last four digits are indeed 2025. So the smallest such n is 925, and the sum of its digits is $9 + 2 + 5 = \boxed{16}$. □